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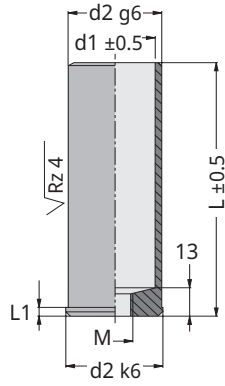
15

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Guiding Components



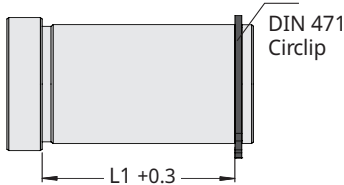
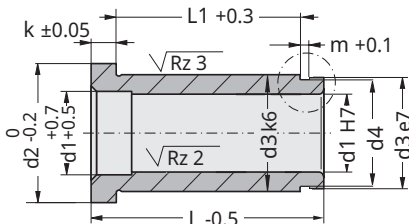
Centring Bush

Code: **G28**

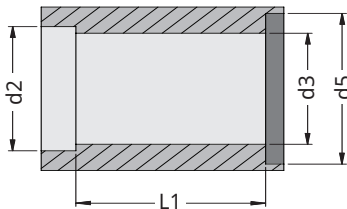
d1	L	d2	M	L1
20	30	16	M8	2.5
	40			
	60			
	80			
	100			
	120			
	140			
	160			
26	30	21	M10	2.5
	40			
	60			
	80			
	100			
	120			
	140			
	160			
30	180	25	M12	2.5
	40			
	60			
	80			
	100			
	120			
	140			
	160			
	180			
	200			
240				



Guide Bush - Headed (with circlip slotted) **Code: G49**



Mounting Dimensions



Order: **G49**. d1 x L



Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

Compatible Pillars



For general usage in injection mould guide systems - it prevents the bush protrude in free rotating plates (H6 centre fixing).

d1	L	d3	d2	k	L1	m	d4	d5	Circlip
12	27	18	25	6	18	1.3	17	27	18
	36				27				
	46				37				
	56				47				
14	27	20	25	6	18	1.3	19	30	20
	36				27				
	46				37				
	56				47				
	66				57				
	76				67				
	86				77				
	96				87				
	116				107				
	16				27				
36		27							
46		37							
56		47							
18	27	26	31	6	18	1.3	25	36	26
	36				27				
	46				37				
	56				47				
	66				57				
	76				67				
	86				77				
	96				87				
	116				107				
	136				127				
22	27	30	35	6	17	1.6	29	41	30
	36				26				
	46				36				
	56				46				
	66				56				
	76				66				
	86				76				
	96				86				
	116				106				
	136				126				
156	146								

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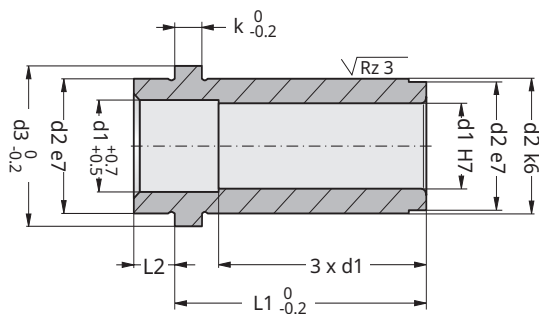
18

Guiding Components



Code: **G31**

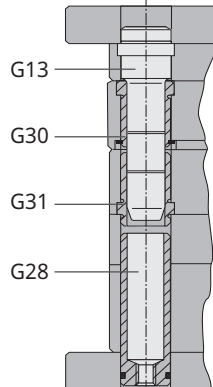
Guide Bush with Centring Collar



d1	L1	d2	d3	k	L2
14	17	20	25	6	9
	22				
	27				
	36				
	46				
	56				
	66				
	76				
	86				
	96				
15	116				
	17	26	31	6	9
	22				
	27				
	36				
	46				
	56				
	66				
	76				
	86				
	96				
18	106				
	116				
	136				

d1	L1	d2	d3	k	L2
22	17	30	35	6	9
	22				
	27				
	36				
	46				
	56				
	66				
	76				
	86				
	96				
24	106				
	116				
	136				
	156				
	27	42	47	6	9
	36				
	46				
	56				
	66				
	76				
	86				
	96				
	106				
	116				
30	126				
	136				
	156				
	196				
32	46	54	60	10	12
	56				
	66				
	76				
	86				
	96				
	116				
	136				
	156				
	196				
40	246				
	46	54	60	10	12
	56				
	66				
	76				
	86				
	96				
42	116				
	136				
	156				
	196				
	246				

Mounting Example



3D CAD
FILES
GUVENAL



Download
www.guvenal.net

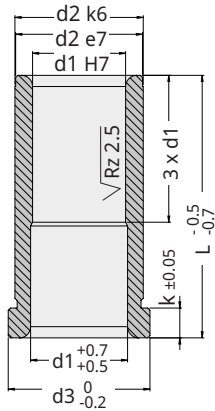
3D CAD - Formats: CATIA, STEP, IGES 3D, X_T



Order: **G31**. d1 x L1



Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC



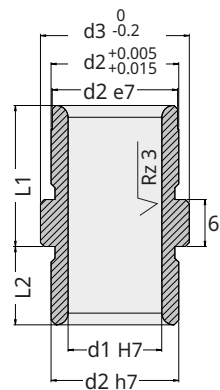
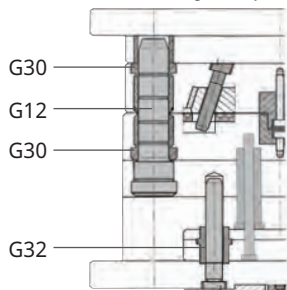
Guide Bush - Headed

Code: **G30**

d1	L	d2	d3	k
10	17	14	16	3
	27			
	36			
	46			
	56			
12	26	16	19	4
	36			
	46			
	56			
	66			
14	26	20	25	6
	36			
	46			
	56			
	66			
16	26	24	28	6
	36			
	46			
	56			
	66			
18	26	26	31	6
	36			
	46			
	56			
	66			
20	26	28	32	6
	36			
	46			
	56			
	66			

d1	L	d2	d3	k
25	26	34	38	8
	36			
	46			
	56			
	66			
30	76	39	43	10
	86			
	96			
	106			
	116			
40	46	54	58	10
	56			
	66			
	76			
	86			
	96			
	106			
	116			
	126			
	136			
	146			

Mounting Example

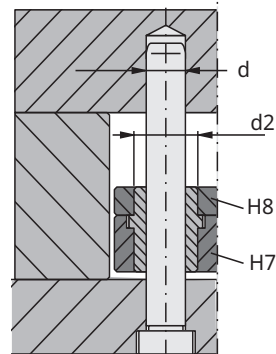


Code: **G32**

Guide Bush with Centre Collar

d1	L1	d2	d3	L2
12	16	16	19	12
14	16	20	25	12
16	16	24	28	12
18	16	26	31	12
20	16	28	32	12
24	21	34	38	16
25	21	34	38	16
30	21	39	43	16

Mounting Example



G32 Bushes ensuring the movements of ejector plates in injection moulds be rigid and precise.



Order: **G32**. d1 x L1



Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC



Order: **G30**. d1 x L



Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

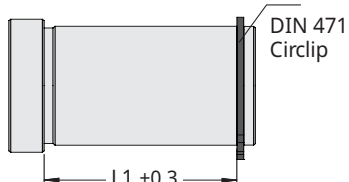
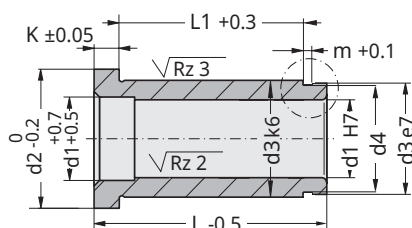


for Turkish Mould
Standard

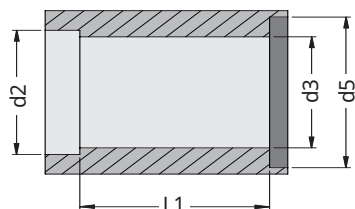


Guide Bush - Headed
(with circlip slotted)

Code: **G77**



Mounting Dimensions



Order: **G77**. d1 x L



Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

Compatible Pillars



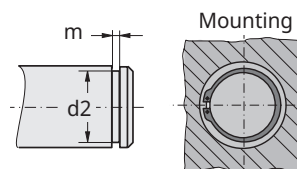
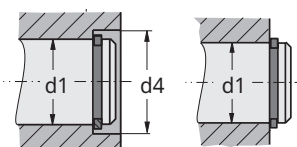
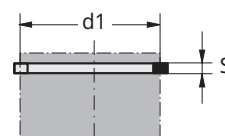
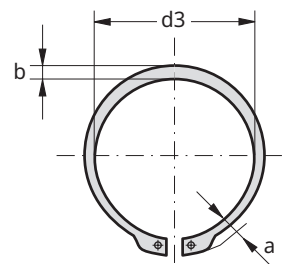
For general usage in injection mould guide systems - it prevents the bush protrude in free rotating plates (H6 centre fixing).

d1	L	d3	d2	k	L1	m	d4	d5	Circlip
12	26				16				
	36				26				
	46	16	19	4	36	1.1	15.2	25	16
	56				46				
	66				56				
14	26				16				
	36				26				
	46	20	25	6	36	1.3	19	30	20
	56				46				
	66				56				
	76				66				
16	26				16				
	36				26				
	46	24	28	6	36	1.3	22.9	34	24
	56				46				
	66				56				
	76				66				
	86				76				
20	26				16				
	36				26				
	46	28	32	6	36	1.6	26.6	38	28
	56				46				
	66				56				
	76				66				
	86				76				
	96				86				
25	26				14				
	36				24				
	46				34				
	56				44				
	66	34	38	8	54	1.6	32.3	45	34
	76				64				
	86				74				
	96				84				
	106				94				
30	36				21				
	46				31				
	56				41				
	66	39	43	10	51	1.8	37	51	39
	76				61				
	86				71				
	96				81				
	106				91				
	116				101				
40	46				28				
	56				38				
	66				48				
	76				58				
	86				68				
	96	54	58	10	78	2.1	51	67	54
	106				88				
	116				98				
	126				108				
	136				118				
	146				128				



External Circlip

Code: **DIN471**



d1	S	a	b	d2	d3	d4	m
16	1	3.5	2.2	15.2	14.7	24.4	1.1
20		4	2.6	19	18.5	29	
21		4.1	2.7	20	19.5	30.2	
22		4.2	2.8	21	20.5	31.4	
24	1.2	4.4	3	22.9	22.2	33.8	1.3
25		4.4	3	23.9	23.2	34.8	
26		4.5	3.1	24.9	24.2	36.1	
28		4.7	3.2	26.6	25.9	38.4	
29		4.8	3.4	27.6	26.9	39.6	
30	1.5	5	3.5	28.6	27.9	41	1.6
34		5.4	3.8	32.3	31.5	45.8	
37		5.7	4.1	35	34.2	49.4	
39	1.75	5.8	4.2	37	35.2	50.6	1.85
45		6.7	4.7	42.5	41.5	59.4	
50	2	6.9	5.1	47	45.8	64.8	2.15
54		7.1	5.3	51	49.8	70	



Order:
DIN471. d1 x S

"Special Series for Injection Moulds"

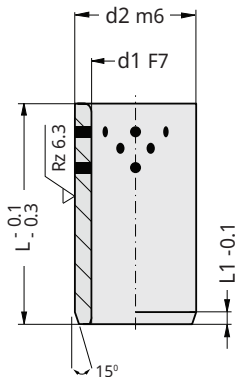


Guide Bush - Plain
Self-Lubricating

Code: **G48**

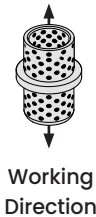
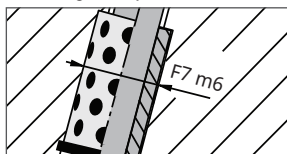
* Hole tolerance for shrink fit: H7

d1	d2	L	L1
8	12	8	2
		10	
		15	
10	14	10	2
		12	
		15	
12	18	20	2
		25	
		30	
		35	
		40	
		45	
15	22	50	2
		55	
		60	
		65	
16	24	70	4
		75	
		80	
18	28	85	4
		90	
		95	
20	30	100	4
		105	
		110	
24	33	115	4
		120	
		125	
25	35	130	4
		135	
		140	
35	38	145	4
		150	
		155	
30	40	160	4
		165	
		170	
40	45	175	4
		180	
		185	
38	50	190	4
		195	
		200	
42	55	205	4
		210	
		215	
45	60	220	4
		225	
		230	



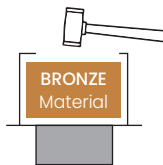
d1	d2	L	L1
48	60	30	4
		40	
		50	
50	62	60	4
		70	
		80	
60	74	100	4
		110	
		120	
80	96	130	4
		140	
		150	
100	120	160	4
		170	
		180	
120	140	190	4
		200	
		210	
150	170	220	4
		230	
		240	
160	180	250	4
		260	
		270	

Mounting Example

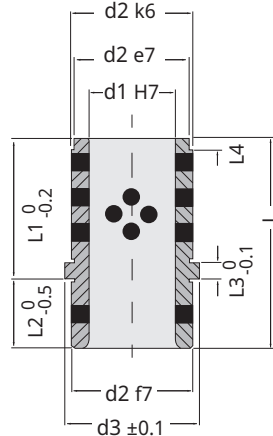


Guide Bush with Centre Collar
Self-Lubricating

Code: **G26**



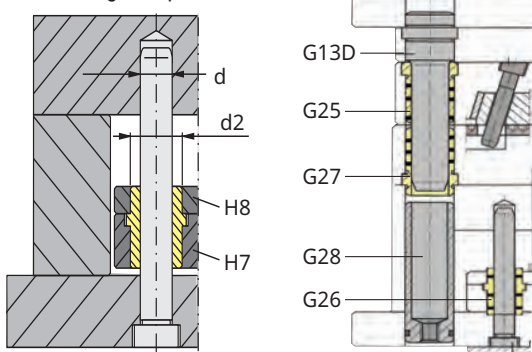
Positioning :
Pls. do not make
direct fitting.
Precision assembly
recommended
with bronze / brass.



d1	L	d2	d3	L1	L2	L3	L4
10	21	14	17	12	9	6	2
12	26	16	19	17	9	6	2
14	26	20	25	17	9	6	2
16	26	20	25	17	9	6	2
20	39	26	31	22	17	6	2
22	49	30	35	27	22	6	4
25	49	30	35	27	22	6	2
30	63	42	47	36	27	6	2
40	63	50	60	36	27	8	6
50	92	63	72	55	37	8	6

Especially, it can be used in pillar systems of the ejector plates in injection moulds with Guide Pillar without oil groove (G17) Guide bushes are self-lubricating systems.
Heat Resistance: 150° C. It can even work in humid environments.

Mounting Example



Order:
G48. d1 x d2 x L



Material:
Bronze + graphite inserts



Order:
G26. d1 x L



Material:
Bronze + graphite inserts

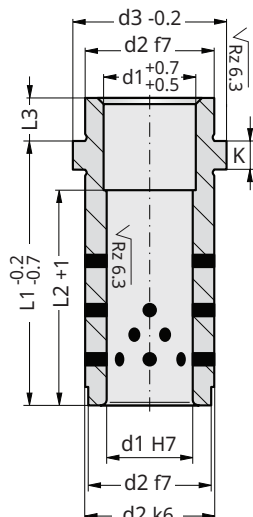
"Special Series for Injection Moulds"



Working
Direction

Guide Bush with Collar
Self-Lubricating

Code: **G27**



G27 products used in injection moulds with **G13D** guide pillars (without oil groove). Higher stability is achieved with centering of the mould in length. This way, a long-term and precision working environment with other equipment (ejector - core etc.) **H7** tolerance is recommended in guide pillar / bush systems. It is compatible with the sudden movements during working. These products are not affected by vibration / impacts.

Heat resistance: 150⁰ c.

d1	L1	d2	d3	K	L3
14	17	20	25	6	6
	22				
	27				
	36				
15	46				
	56				
	L2: > 2 - 3 x d1				
16	17	20	25	6	8
	22				
	27				
	36				
	46				
	56				
	66				
	76				
L2: > 2 - 3 x d1					
20	17	26	31	6	8
	22				
	27				
	36				
	46				
	56				
	66				
	76				
	86				
	96				
18	L2: > 2 - 3 x d1				

d1	L1	d2	d3	K	L3
25	22	30	35	6	8
	27				
	36				
	46				
56					
66					
76					
86					
22	96	L2: > 2 - 3 x d1			
24	116				
30	27	42	47	6	8
	36				
	46				
	56				
	66				
	76				
	86				
	96				
116	L2: > 2 - 3 x d1				
136					
32	136				
40	46	54	60	10	12
	56				
	66				
	76				
	86				
	96				
	116				
	136				
156	L2: > 2 - 3 x d1				
196					
42	196				

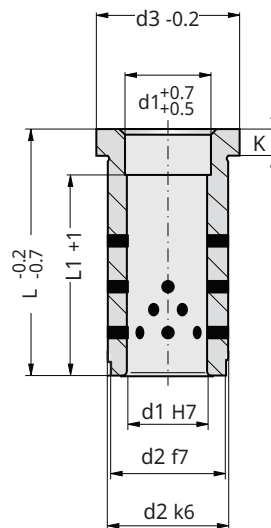
Sliding bushing systems provide higher stability.
These products are lubricated once as a thin layer and this reduces negative effects.



Working
Direction

Guide Bush Headed
Self-Lubricating

Code: **G25**



G25 products used in injection moulds with **G12D** and **G50D** guide pillars (without oil groove).

H7 tolerance is recommended in guide bush systems. It is compatible with the sudden movements during working. These products are not affected by vibration / impacts.

Heat resistance: 150⁰ c.
It can even work in humid environments.

d1	L	d2	d3	K			
12	16	18	22	4			
	21						
	27						
	36						
		L1: 2 - 3 x d1					
16	27	22	27	6			
	36						
15	46				L1: 2 - 3 x d1		
	56						
20	27	26	31	6			
	36						
	46						
	56						
18	66	L1: 2 - 3 x d1					
	76						
	96						
25	36	30	35	6			
	46						
	56						
22	66				L1: 2 - 3 x d1		
	76						
24	86						
	96						

d1	L1	d2	d3	K
30 32	36	42	47	6
	46			
	56			
	66			
	76			
	86			
	96			
	116			
L1: 2 - 3 x d1				
40 42	36	54	60	10
	46			
	56			
	66			
	76			
	86			
	96			
	116			
136				
L1: 2 - 3 x d1				
50	76	66	72	10
	86			
	96			
	116			
	136			
	156			
196				
L1: 2 - 3 x d1				
60	96	80	86	20
	116			
	136			
	156			
	196			
L1: 2 - 3 x d1				



Order:
G27. d1 x L1



Material:
Bronze + graphite inserts

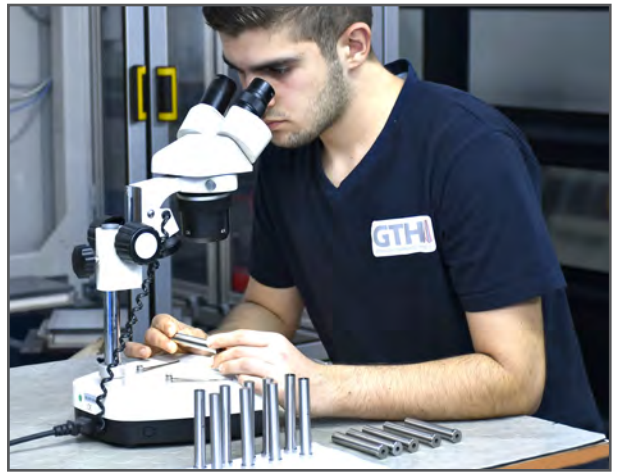


Order:
G25. d1 x L



Material:
Bronze + graphite inserts

Stable and reliable Die & Mould production is obtained with components having a reliable quality. Güvenal tends towards the highest quality standards with all development and test efforts.



Güvenal: Reliability and Excellence in the Tooling Industry Güvenal is a well-known manufacturer in the industry, recognized for its wide product range and fast distribution systems. Güvenal's meticulous Quality and Control Processes ensure the highest quality at every stage of the production process. Strict controls are applied at each step, from raw material selection to the final product formation. To ensure precision shaping quality, parameters such as circularity, concentricity, flatness, and perpendicularity are carefully monitored through 3D inspection technology. Additionally, hardness tests, along with hardened and induction parameter records, are conducted at our facility. Thanks to these processes, we continue to offer a wide variety of high-quality, excellent products with the lowest tolerances under the Güvenal Quality Assurance Labels.

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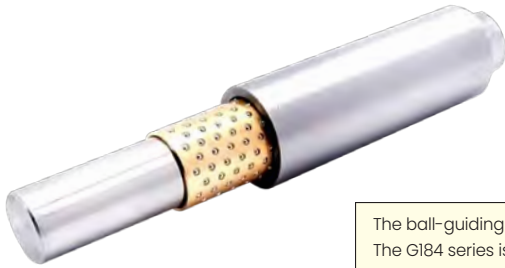
16

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Guiding Components

NEW



Code: **G184-1**

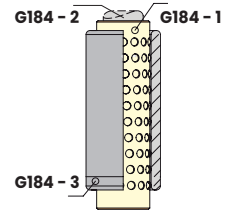
MINI Ball Cage

Code: **G184-2**

MINI Guide Pillar

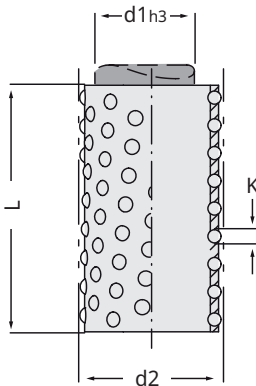
Code: **G184-3**

MINI Guide Bush for G184-1



The ball-guiding mini-series enables achieving maximum precision with smooth sliding.

The G184 series is widely used in high-speed stamping, injection moulding, control instruments, automation, robotics, and the semiconductor industries. Additionally, we can provide ball cages, bushes, and pillars made to custom specifications based on drawings.



Code: **G184-1**

MINI Ball Cage

Material: Bronze Cage B14

Balls: Steel for bearings 100Cr6

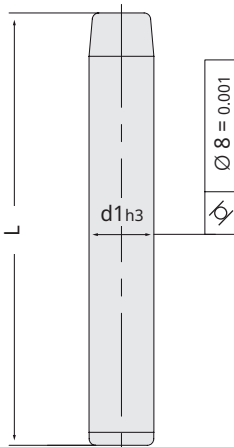
Superfinished grade AAA



Order:
G184-1. d1 x L

d1	L	d2	K	No. of Balls
4	10	6	1	28
	15			42
	20			63
	25			77
5	10	7	1	32
	15			48
	20			72
	25			88
	30			112

d1	L	d2	K	No. of Balls
6	10	8	1	36
	15			54
	20			81
	25			99
8	30	10	1	126
	15			60
	20			90
	25			110
	30			140
	40			190



Code: **G184-2**

MINI Guide Pillar
DIN 9825 / ISO 9182

Material: 1.7131

Hardness: 58-62 HRC

Heat treatment: Case-hardening and tempering

Note: Fine grinding and super finishing

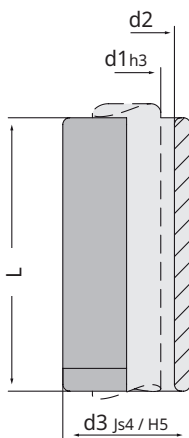


Order:
G184-2. d1 x L

d1	L
4	40
	50
	60
	80
	100
5	40
	50
	60
	80
	100

d1	L
6	40
	50
	60
	80
	100
8	100
	125
	140
	160
	160

d1	L
6	50
	60
	80
	100
	125
8	125
	140
	160
	160
	160



Code: **G184-3**

MINI Guide Bush for Ball Cage

Material: 1.7131

Hardness: 60-64 HRC

Heat treatment: Case-hardening and tempering

Note: Fine grinding and super finishing



Order:
G184-3. d1 x L

d1	d2	d3	L
4	6	8	10
			15
			20
			25
5	7	10	25
			30
			35
			40
			40

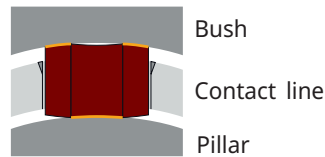
d1	d2	d3	L
6	8	11	15
			20
			25
			30
8	10	14	35
			40
			40
			40
			40

NEW

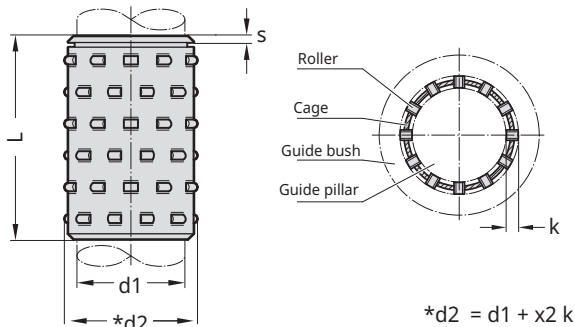
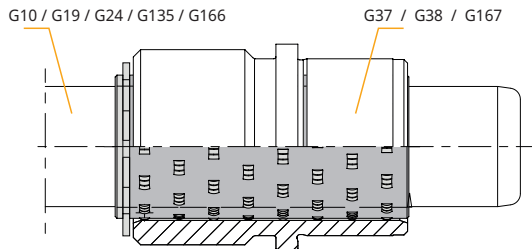


Roller Cage Code: **G81**

G81 Roller Cages create a linear contact with the guide bush and guide pillar, resulting in a significantly higher load-carrying capacity for each roller compared to a ball-cage of the same diameter. The profile rollers are arranged in a spiral pattern along the axis, ensuring each roller moves along its own unique path. The cages are designed with grooves to accommodate a DIN 471 circlip.



- Ideal for high-speed operations
- Suitable for applications requiring maximum precision
- Offers greater stiffness compared to ball cages
- Well-suited for **Fine Blanking & Progressive Dies** due to its ability to withstand high shear forces
- Suitable for **IML Injection Moulds**
- Reduced wear due to the axial offset of the rollers
- Optimized inlet geometry ensures smooth roller entry
- Play-free guidance achieved through optimal preload
- Can be used for axial motions only



d1	L	No. of Rollers	k	s
20	45	32	3	1.6
	55	40		
	65	48		
	75	56		
	85	64		
	95	72		
25	105	80	3	1.6
	45	40		
	55	50		
	65	60		
	75	70		
	85	80		
30	95	90	4	2.1
	105	100		
	115	110		
	125	120		
	45	48		
	55	60		
30	65	72	4	2.1
	75	84		
	85	96		
	95	108		
	105	120		
	115	132		
30	125	144	4	2.1
	135	156		
	145	168		
	155	180		
	165	192		
	175	204		

d1	L	No. of Rollers	k	s
40	55	70	4	2.1
	65	84		
	75	98		
	85	112		
	95	126		
	105	140		
40	115	154	4	2.1
	125	168		
	135	182		
	145	196		
	155	210		
	165	224		
50	175	238	4	2.1
	185	252		
	205	280		
	65	108		
	75	126		
	85	144		
50	95	162	4	2.1
	105	180		
	115	198		
	125	216		
	135	234		
	145	252		
63	155	270	4	2.1
	165	288		
	175	306		
	185	324		
	205	360		
	75	154		
63	85	176	4	2.1
	95	198		
	105	220		
	115	242		
	125	264		
	135	286		
63	145	308	4	2.1
	155	330		
	165	352		
	175	374		
	185	396		
	205	440		

Roller Cage: Aluminium

Rollers: Steel hardened, 100 Cr6, DIN 5402

Order: **G81**. d1 x L

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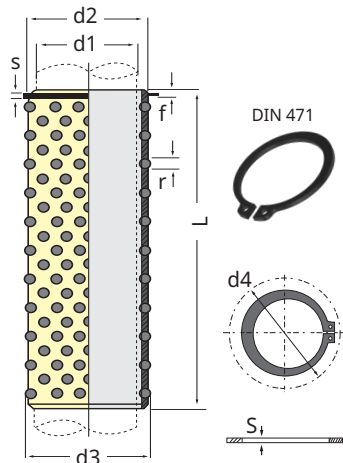
Guiding Components



Ball Cage

Code: **G70**

Heat resistance of ball cage is 120° C.
Ball cages are presented with circlips.



Mounting Example



Compatible Bushes



$$L = K + 0.5 \times s$$

d1	L	d2	Ball		Circlip		
			d3	r	d4	f	s
NEW 10	30	13	14	2	20.6	3.2	1.2
	40						
12	40	15	16	2	22.6	3.2	1.2
	56						
15	45	20	21	3	28.4	5	2
	65						
16	71	21	22	3	29.6	5	2
18	45	23	24	3	32.2	5	2
	56						
	71						
19	45	24	25	3	33.2	5	2
	56						
20	71	25	26	3	34.2	5	2
	80						
	95						
24	45	29	30	3	39.1	5	2
	56						
25	71	30	31	3	40.5	5	2
	80						
	95						
	105						
	120						
30	45	39	40	4	49	5	2
	56						
32	75	37	38	4	51.4	5	2
	80						
	95						
	105						
	120						
	140						

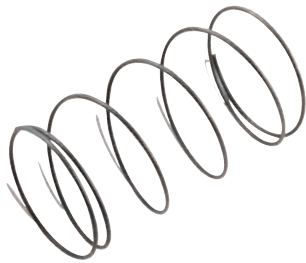
d1	L	d2	Ball		Circlip		
			d3	r	d4	f	s
	56						
	71						
	80						
38	95	45	46	4	59.1	5	2
40	105	47	48		60.8		
	120						
	140						
	160						
	71						
	80						
48	95	55	56	4	70.2	5	2
	105						
50	120	57	58		72.6		
	140						
	160						
	85						
	95						
60	105	67	68	4	83.1	5.7	2.7
63	120	70	71		87		
	140						
	160						
	105						
	120						
80	140	90	92	6	108.5	6.7	3.2
	160						
	200						



Order: **G70**. d1 x L



Material: Ms58 **Ball:** 100Cr6
Tolerance: +0.001



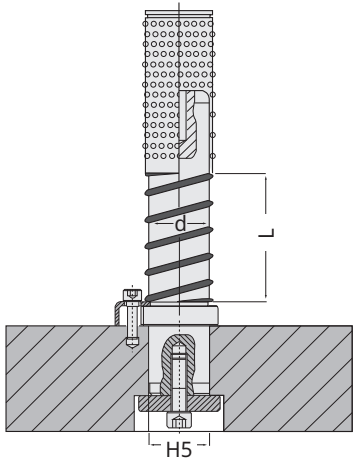
Code: **G70 - 1**

Ball Cage Retaining Spring

Note: Production in ball systems or other die inner designs Upon request.
L: 10 mm ranges

d	L
19 20	Up to 40 ~ 140
24 25	Up to 40 ~ 180
30 32	Up to 50 ~ 230
38 40	Up to 60 ~ 280
48 50	Up to 70 ~ 280
60 63	Up to 80 ~ 250

Order:
G70-1. d x L



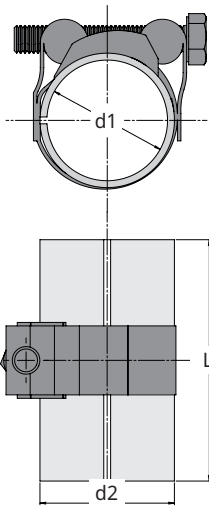
To avoid hanging at usage of ball cage bush, ball cage retaining spring can be used.



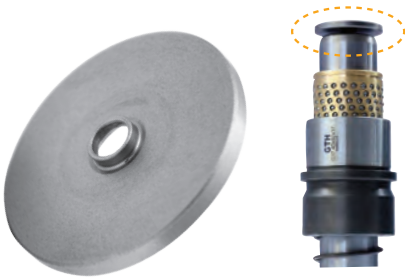
Limiting Bush

Code: **G168**

Order Code	d1	Guide Pillar	d2	L
G16816	16	15 / 16	20	20
G16820	20	19 / 20	30	20
G16825	25	24 / 25	30	30
G16832	32	30 / 32	40	30
G16840	40	38 / 40	50	40
G16850	50	48 / 50	60	40
G16863	63	60 / 63	70	40



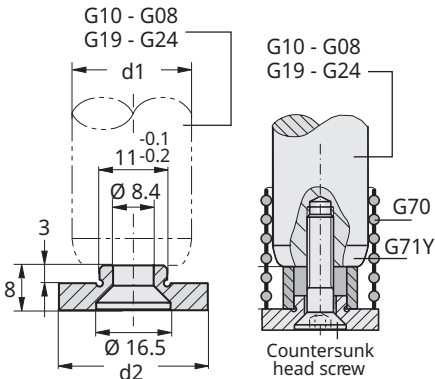
It helps to position the working length of the bushes running on the guide pillars.



Code: **G151**

Ball Cage Holder Flange

(Locking flange for ball cage)



d1	d2	Pillar Dia.
20	25	Ø 20
25	30	Ø 25
30	37	Ø 30
40	47	Ø 40
50	57	Ø 50
60	67	Ø 60
63	71	Ø 63

Order: **G151. d1**

Material: 1.7131 (16MnCr5)

Holder Flange / (locking flange for ball cage)
In ball cage bush systems, it prevents detaching of cage (**G70**) from guide pillar during operating. It is practical and provides convenience in die mounting. **In mounting to die:** Spacer tube (**G71Y**) of ball cage stroke distance and countersunk head screw (**HIC**) can be adjusted in desired length.

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- 17
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Guiding Components

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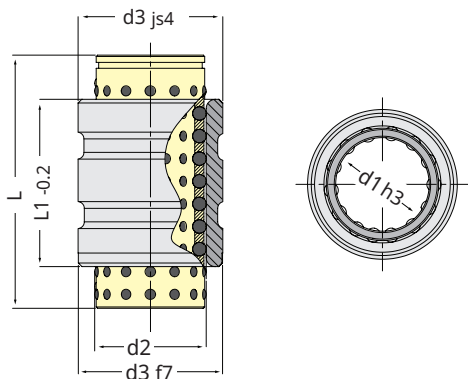
17

18



Plain Guide Bush for Ball Cage
ISO 9448-3

Code: **G167**



d1	d2	d3	L1	L
NEW 10	14	22	23	30
			30	40
			37	40
12	16	22	23	40
			30	40
			37	40
15 16	21 22	28	23	45
			30	45
			37	45
			47	56
			60	71

d1	d2	d3	L1	L
19 20	25 26	32	23	45
			30	45
			37	45
			47	56
			60	71
24 25	30 31	40	77	95
			23	45
			30	45
			37	45
			47	56
30 32	38 40	48	60	71
			77	95
			95	120
			30	45
			37	50
38 40	46 48	58	47	56
			60	71
			77	95
			95	120
			120	140
48 50	56 58	70	30	45
			37	50
			47	63
			60	80
			77	95
60 63	68 71	85	95	120
			120	140
			60	95
			77	95
80	92	105	95	120
			120	140

Note: Ball Cage "G70" must be ordered separately.



Order:
G167. d1 x L1

Material: 1.7131
Hardness: 60 - 64 HRC



The inner diameter of the products is honed, the outer diameter is precision ground. The grooves on the product are made to provide better adhesion to the hole during mounting. **Advantages of G167:** No problem finding the position when changing the bushes. High accuracy and robustness. **Mounting Instruction:** Please secure with adhesive (Order Code: GL.64832) Do not press-fit.



Adhesive / Loctite 648

The adhesive provides optimum push retention for "G167" bushes. For adhesion of bearings bushes, screws and other machine components. Very fast adhesion to handling strength.

Heat Resistance: -55° / +175° C

Package: 50ml / 55 g.



Order: **GL.64832**



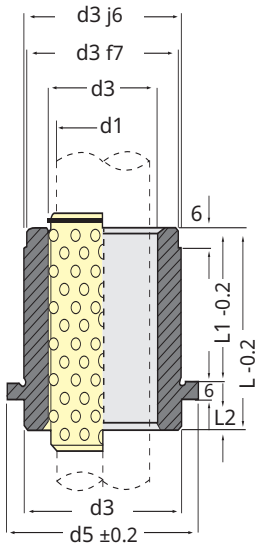
Guide Bush for Ball Cage
(Short type)

Code: **G38**



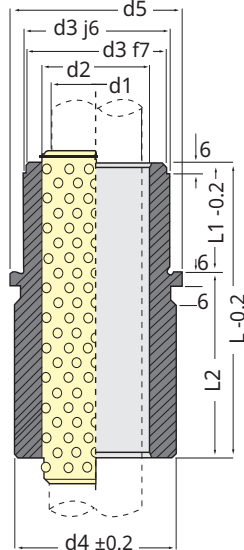
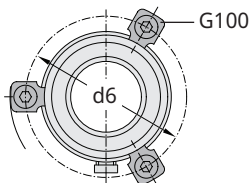
Guide Bush for Ball Cage
(Long type)

Code: **G37**



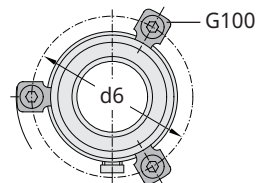
G38 guide bushes work without problem at dies, cuttings under 0.5 mm sheet metal thickness and where cutting space is low and when 200 stamp per minute are exceeded. It is not resistant to the lateral loads. They are preferred in cases that precision should be increased.

4 pcs. clamp for **d1: 38 mm** and larger sizes.



High precision guide bushes for ball cage: Despite providing high reliability at high speeds, contacting balls from one point, brings the system to precision status against impacts and side loads. This disadvantage is compensated to give product diameters up to a certain point in the upper tolerance, the working surface should be precise.

4 pcs. clamp for **d1: 38 mm** and larger sizes.



d1	L	d2	d3	d5	d6	L1	L2
15	35	21	30	34	46.2	23	12
16		22					
19	35	25	32	40	52.4	23	12
20		26					
24	35	30	40	48	60.8	23	12
25		31					
30	42	37	48	56	67.8	30	12
32		40					
38	52	46	58	66	78	37	15
40		48					
48	65	56	70	80	93.2	47	18
50		58					
60	80	68	85	95	108.4	60	20
63		71					
80	80	92	105	118	131.5	60	20



Order:
G38. d1 x L x L1

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

d1	L	d2	d3	d4	d5	d6	L1	L2
15	43	21	30	32	34	46.2	23	20
16		22						36
19	43	25	32	39	40	52.4	23	20
20		26						36
24	59	30	40	46	48	60.8	23	36
25		31						56
30	75	38	48	53	56	67.8	30	45
32		40						63
38	82	46	58	63	66	78	37	45
40		48						71
48	97	56	70	77	80	93.2	47	50
50		58						80
60	116	68	85	92	95	108.4	60	56
63		71						90
80	120	92	105	115	118	131.5	60	60
	150							90



Order:
G37. d1 x L x L1

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

Note: Two different products in section (d1) are to avoid reverse closing of dies during mounting by using three pcs. main dimensions and one pcs. auxiliary (**d1= Ø15 - 19 - 24 - 32 - 38 - 48 - 60 mm**) dimensions at same die, is avoided reverse closing of die plate. Thus, symmetrical die closure is ensured. **Example:** In the same die (Ø20 x 3 - Ø19 x 1) = 4 pieces **Note:** Should be ordered with G70.



interior view



Linear Ball Bearing Bush

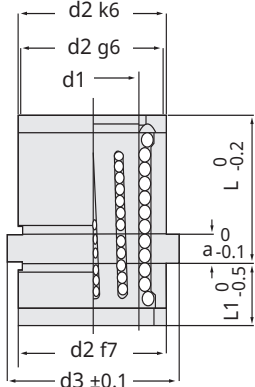
Code: **G75**

Free stroke, rapid and limitless movement.
New for ejector plate.

Operating Temp.: Max. 200°



Positioning :
Pls. do not make
direct fitting.
Precision assembly
recommended
with bronze / brass.



d1	L	d2	d3	a	L1	Ball Dia.	Line Pcs.
12	22	24	28	6	8	3	5
18	26	30	34	6	9	3	6
20	26	32	36	6	9	3	6
	36						
25	26	40	45	6	10	3	8
	36						
	46						
30	46	48	54	8	12	4	8
	66						
40	46	60	66	8	12	4	8
	66						
50	56	70	76	10	25	4	10



Order: **G75**. d1 x L

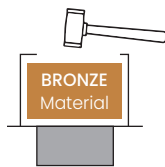


Linear Ball Bearing Bush
Headed

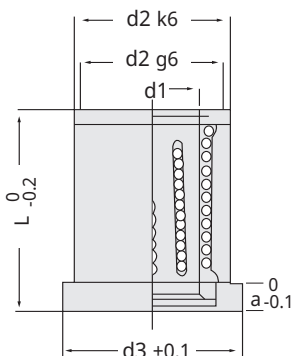
Code: **G74**

Balls are worked independently by rotating,
they are not frictional.

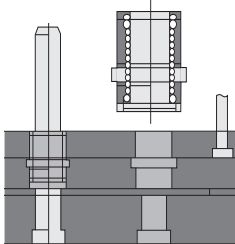
Operating Temp.: Max. 200°



Positioning :
Pls. do not make
direct fitting.
Precision assembly
recommended
with bronze / brass.



d1	L	d2	d3	a	Ball Dia.	Line Pcs.
18	35	30	34	6	3	6
20	35	32	36	6	3	6
	45					
	55					
30	45	48	54	8	4	8
	63					
40	45	60	66	8	4	8
	63					



Order: **G74**. d1 x L



New Linear Ball Bearing Bushes G74 - G75 - G149 - G175: There are balls at inclined - serial - parallel intermediate channels - lines inside of linear **G74-G75-G149-G175** bushes (balls move by rotating in slot and frictionless). In this way, axial motion remains limitlessly free. Also, bearing logic is provided to the steel bush. **The advantage of our new product G75 according to other similar product G29:** Extra stroke distances at unlimited axial direction and robust powerful centering is provided - free motion unlimited with low tolerance absorbs other pinking and tauting. It is presented unlimited motion advantages with compatibility to high speeds.
Linear bush G74: Quality control equipment - practical and reliable product for machine and mould production and device design.
For maintenance and cleaning: "Güvenal Signum (order code: W160110)" spray oil is used.



WARNING ! During production stage of **G74 - G75 - G149 - G175** products ball bearings are positioned at a certain angle. Other imitation products has no angle and the balls are at a single centre.

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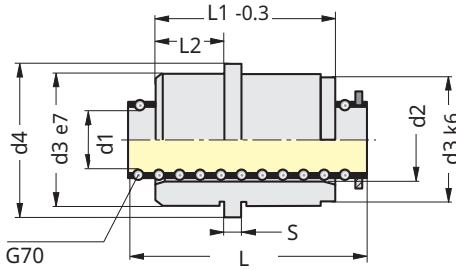
18

Guiding Components



Steel Bush + Ball Cage Bush Set

Code: **G29**

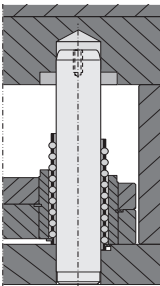


d1	L	L1	L2	d2	d3	d4	S	H
12	40	24	6	16	22	26	6	50
	56							82
18	45	34	11	24	30	35	6	44
	56							66
	71							96
30	56	54	21	38	46	52	6	32
	75							70
	95							110

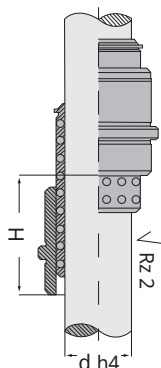


It provides more precision and rapid motion at ejector plates in injection moulds.

Mounting Example



more
precision
and rapid
motion



Order: **G29**. d1 x L



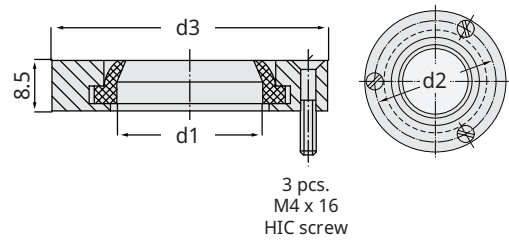
Ball Material: 1.3505 (100Cr6)

Cage: Ms58 Steel Bush: 1.7131 (16MnCr5)



Perforated Flange

Code: **G22**



Order Code	d1	d2	d3
G22 24	24	45	55
G22 25	25	45	55
G22 32	32	55	65
G22 30	30	55	65
G22 38	38	65	75
G22 40	40	65	75
G22 48	48	78	94
G22 50	50	78	94
G22 60	60	92	110
G22 63	63	92	110



Material: 1.7131 (16MnCr5)



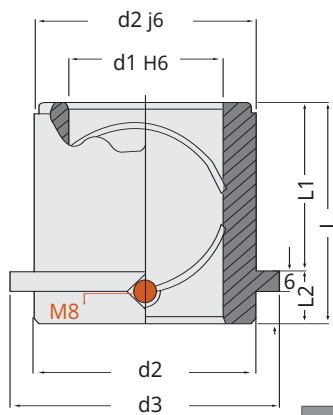
Guide Pillar Protection Flange:

Guide pillar protection flange protects dirt to be occurred at dies or thrust etc. Different various usage areas can be created in dies.

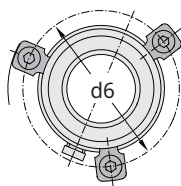


Guide Bush, Thick & Short Type
(with collar & lubricating)

Code: **G34K**

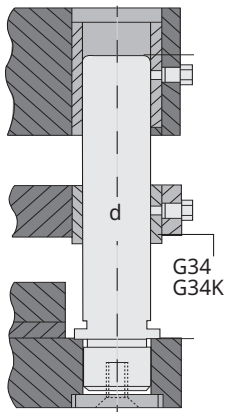


G100
Clamping Detail



4 pcs. clamp for
d1 = 38 mm and
larger sizes.

Mounting Example



d1	L	d2	d3	L1	L2
19	35	32	40	23	12
20					
24	35	40	48	23	12
25					
32	42	48	56	30	12
30					
38	52	58	66	37	15
40					
48	65	70	80	47	18
50					
60	80	85	95	60	20
63					
80	80	105	118	60	20



Order:
G34K. d1 x L x L1

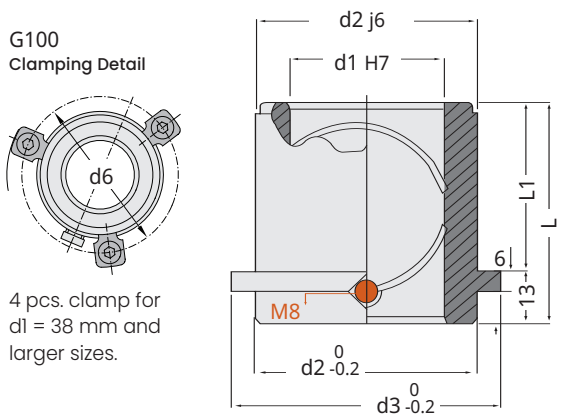
Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

Thick type steel bushes: To provide perpendicularity at bush and guide pillar with new design of **Güvenal Die Components** on product suitable flange fixation (balanced layout) on bearing area is created. During mounting, shrink fit should not be done. To avoid clamping and surface disorders at products by inserting soft tight products to the slots, mounting can be ensured with clamps.

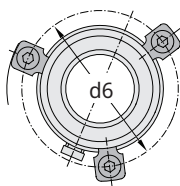


Guide Bush, Thin & Short Type
(with collar & lubricating)

Code: **G34**



G100
Clamping Detail



4 pcs. clamp for
d1 = 38 mm and
larger sizes.

Mounting Details:

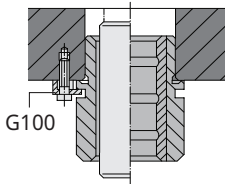
Code: **G34K**

Code: **G34**

d1	d6
20	52.4
25	60.8
30	69
40	79
50	93.2
63	108.4
80	131.5

d1	d6
16	46.2
20	46.2
25	51.4
30	56.4
40	66.8
50	77

d1	L	d2	d3	L1
16	36	28	34	23
20	36	28	34	23
25	36	34	39	23
30	50	39	44	37
40	50	50	54	37
50	60	60	64	47



Order:
G34. d1 x L x L1

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

For being **Güvenal Die Components** products having long life and maximum resistance against abrasion, unique precision has provided at surface roughness. This is also created specific bushing system.

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Guiding Components

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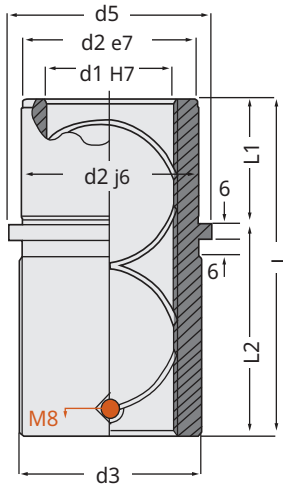
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Guiding Components

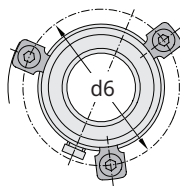


Guide Bush, Thin & Long Type (with collar & lubricating)

Code: **G33**



G100
Clamping Detail



4 pcs. clamp for
d1 = 38 mm and
larger sizes.

Mounting Details:

Code: **G33**

Code: **G33K**

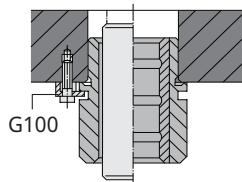
d1	L	d2	d3	L1	L2
16	70	28	34	23	47
20	70	28	34	23	47
25	80	34	39	23	57
30	90	39	44	37	53
40	100	50	54	37	63
50	120	60	64	47	73

d1	d6	d1	d6
16	46.2	20	52.4
20	46.2	25	60.8
25	51.4	30	69
30	56.4	40	79
40	66.8	50	93.2
50	77	63	108.4
		80	131.5



Order:
G33 d1 x L x L1

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

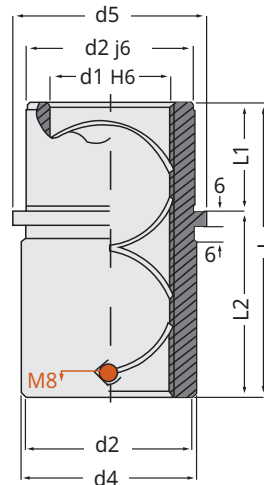


For being **Güvenal Die Components** products having long life and maximum resistance against abrasion, unique precision has provided at surface roughness. This is also created specific bushing system.

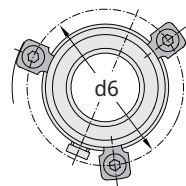


Guide Bush, Thick & Long Type (for heavy dies / with collar & lubricating)

Code: **G33K**

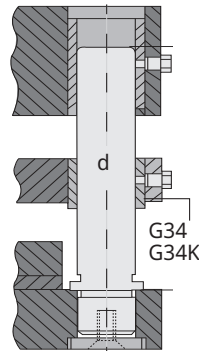


G100
Clamping Detail



4 pcs. clamp for
d1 = 38 mm and
larger sizes.

Mounting Example



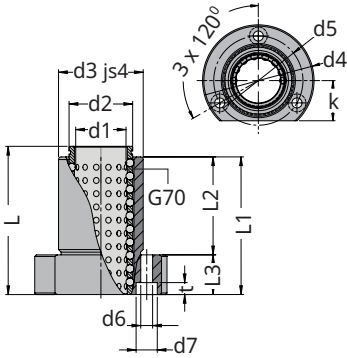
Order:
G33K d1 x L x L1

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

d1	L	d2	d4	d5	L1	L2
19	43	32	39	40	23	20
20	59					36
24	59	40	46	48	23	36
25	79					56
32	75	48	53	56	30	45
30	93					63
38	82	58	63	66	37	45
40	108					71
48	97	70	77	80	47	50
50	127					80
60	116	85	92	95	60	56
63	150					90
80	120	105	115	118	60	60
	150					90



Thick type steel bushes: Please don't use two products without oil groove together. One of product in guide pillar or bushes should be used as set with oil groove. While using guide pillar and bush sets, should be considered that both of them have same brand. Because, grinding tolerance can be different in different guide pillars and bushes. Also, this can be caused big problems during both mounting and operating. The dimensions not giving their tolerance are subjected to general tolerance (DIN 7168).



Code: **G72 - 1**

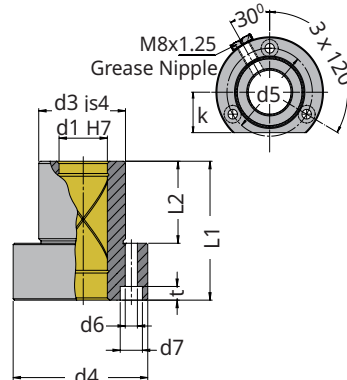
Guide Bush with Flange, for Ball Cage

- * 3 pcs "IMB" coded screws are used to fixed product.
- * Clamping Screws & "G70" coded Ball Cage Bush should be ordered separately.

Order:
G72 - 1. d1 x L1 x L2

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

d1	d2	d3	d4	d5	d6	d7	t	k	L	L1	L2	L3
19	25	32	50	40	4.5	8	5	18	45	38	23	15
20	26								56	45	30	
									72	52	37	
24	30	40	63	50	5.5	10	6	23	45	38	23	15
25	31								71	55	30	
									72	62	37	25
30	38	48	72	58	5.5	10	6	28	56	45	30	15
32	40								71	62	37	
									80	72	47	25
38	46	58	85	70	7	11	7	33	63	55	30	25
40	48								80	67	37	
									95	77	47	30
48	56	70	104	86	9	15	9	38	80	62	37	25
50	58								95	89	47	
									120	102	60	42
60	68	85	120	100	9	15	9	46	95	89	47	
63	71								120	102	60	42
80	92	105	148	125	12	18	12	56	140	125	75	50



Code: **G72 - 2**

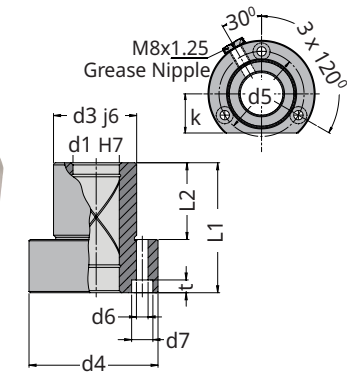
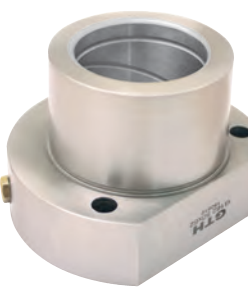
Guide Bush with Flange, Bronze Coated

- * 3 pcs "IMB" coded screws are used to fixed product.
- * Clamping Screws should be ordered separately.

Order:
G72 - 2. d1 x L1 x L2

Material: 1.1213 (Cf53)
Bronze Casting

d1	d3	d4	d5	d6	d7	t	k	L1	L2
19	32	50	40	4.5	8	5	18	38	23
20								45	30
								52	37
24	40	63	50	5.5	10	6	23	38	23
25								55	30
								62	37
30	48	72	58	5.5	10	6	28	45	30
32								62	37
								72	47
38	58	85	70	6.6	11	7	33	55	30
40								67	37
								77	47
48	70	104	86	9	15	9	38	62	37
50								89	47
								102	60
60	85	120	100	9	15	9	46	89	47
63								102	60
80	105	146	125	11	18	12	56	125	75



Code: **G72 - 3**

Guide Bush with Flange, Steel

- * 3 pcs "IMB" coded screws are used to fixed product.
- * Clamping Screws should be ordered separately.

Order:
G72 - 3. d1 x L1 x L2

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

d1	d3	d4	d5	d6	d7	t	k	L1	L2
19	32	50	40	4.5	8	5	18	38	23
20								45	30
								52	37
24	40	63	50	5.5	10	6	23	38	23
25								55	30
								62	37
30	48	72	58	5.5	10	6	28	45	30
32								62	37
								72	47
38	58	85	70	6.6	11	7	33	55	30
40								67	37
								77	47
48	70	104	86	9	15	9	38	62	37
50								89	47
								102	60
60	85	120	100	9	15	9	46	89	47
63								102	60
80	105	146	125	11	18	12	56	125	75

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Guiding Components